AB Testing (Marketing Campaigns)

Import Dependencies

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as st
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score,mean_squared_error
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import coint
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
# Loading dataset
df = pd.read_csv('marketing_campaign.csv')
```

In [3]:
```
# data overview
df.head()
```

Out[3]:

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Fa Con (Conv /
0	01-01-2025	FB_Jan25	2116	18	8	$126	0.83%	
1	01-02-2025	FB_Jan25	3106	36	12	$104	1.15%	
2	01-03-2025	FB_Jan25	3105	26	8	$102	0.84%	
3	01-04-2025	FB_Jan25	1107	27	9	$71	2.45%	
4	01-05-2025	FB_Jan25	1317	15	7	$78	1.10%	

In [4]:
```
# data info
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 365 entries, 0 to 364
Data columns (total 17 columns):
 #   Column                                     Non-Null Count  Dtype
---  ------                                     --------------  -----
 0   Date                                       365 non-null    str
 1   Facebook Ad Campaign                       365 non-null    str
 2   Facebook Ad Views                          365 non-null    int64
 3   Facebook Ad Clicks                         365 non-null    int64
 4   Facebook Ad Conversions                    365 non-null    int64
 5   Cost per Facebook Ad                       365 non-null    str
 6   Facebook Click-Through Rate (Clicks / View)      365 non-null    str
 7   Facebook Conversion Rate (Conversions / Clicks)  365 non-null    str
 8   Facebook Cost per Click (Ad Cost / Clicks)       365 non-null    str
 9   AdWords Ad Campaign                        365 non-null    str
 10  AdWords Ad Views                           365 non-null    int64
 11  AdWords Ad Clicks                          365 non-null    int64
 12  AdWords Ad Conversions                     365 non-null    int64
 13  Cost per AdWords Ad                        365 non-null    str
 14  AdWords Click-Through Rate (Clicks / View)       365 non-null    str
 15  AdWords Conversion Rate (Conversions / Click)    365 non-null    str
 16  AdWords Cost per Click (Ad Cost / Clicks)        365 non-null    str
dtypes: int64(6), str(11)
memory usage: 48.6 KB
```

In [5]:

```
# data columns
df.columns
```

Out[5]:

```
Index(['Date', 'Facebook Ad Campaign', 'Facebook Ad Views',
       'Facebook Ad Clicks', 'Facebook Ad Conversions', 'Cost per Facebook Ad',
       'Facebook Click-Through Rate (Clicks / View)',
       'Facebook Conversion Rate (Conversions / Clicks)',
       'Facebook Cost per Click (Ad Cost / Clicks)', 'AdWords Ad Campaign',
       'AdWords Ad Views', 'AdWords Ad Clicks', 'AdWords Ad Conversions',
       'Cost per AdWords Ad', 'AdWords Click-Through Rate (Clicks / View)',
       'AdWords Conversion Rate (Conversions / Click)',
       'AdWords Cost per Click (Ad Cost / Clicks)'],
      dtype='str')
```

In [7]:

```
# descriptive stats
df.describe()
```

Out[7]:

	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	AdWords Ad Views	AdWords Ad Clicks	AdWords Ad Conversions
count	365.000000	365.000000	365.000000	365.000000	365.000000	365.000000
mean	2209.375342	44.065753	11.758904	4717.739726	60.383562	5.980822
std	620.284443	12.110574	2.981990	561.261203	14.368225	1.628106
min	1050.000000	15.000000	5.000000	3714.000000	31.000000	3.000000
25%	1656.000000	35.000000	10.000000	4252.000000	49.000000	5.000000
50%	2301.000000	43.000000	12.000000	4711.000000	60.000000	6.000000
75%	2717.000000	54.000000	13.000000	5200.000000	73.000000	7.000000
max	3320.000000	73.000000	25.000000	5760.000000	89.000000	9.000000

In [11]:
```
# converting date into date time
df['Date'] = pd.to_datetime(df['Date'], format='mixed')
```

In [12]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 365 entries, 0 to 364
Data columns (total 17 columns):
 #   Column                                      Non-Null Count  Dtype
---  ------                                      --------------  -----
 0   Date                                        365 non-null    datetime64[us]
 1   Facebook Ad Campaign                        365 non-null    str
 2   Facebook Ad Views                           365 non-null    int64
 3   Facebook Ad Clicks                          365 non-null    int64
 4   Facebook Ad Conversions                     365 non-null    int64
 5   Cost per Facebook Ad                        365 non-null    str
 6   Facebook Click-Through Rate (Clicks / View) 365 non-null    str
 7   Facebook Conversion Rate (Conversions / Clicks) 365 non-null    str
 8   Facebook Cost per Click (Ad Cost / Clicks)  365 non-null    str
 9   AdWords Ad Campaign                         365 non-null    str
 10  AdWords Ad Views                            365 non-null    int64
 11  AdWords Ad Clicks                           365 non-null    int64
 12  AdWords Ad Conversions                      365 non-null    int64
 13  Cost per AdWords Ad                         365 non-null    str
 14  AdWords Click-Through Rate (Clicks / View)  365 non-null    str
 15  AdWords Conversion Rate (Conversions / Click) 365 non-null    str
 16  AdWords Cost per Click (Ad Cost / Clicks)   365 non-null    str
dtypes: datetime64[us](1), int64(6), str(10)
memory usage: 48.6 KB
```

In [15]:
```
df.head()
```

Out[15]:

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Fa Con (Conv
0	2025-01-01	FB_Jan25	2116	18	8	$126	0.83%	
1	2025-01-02	FB_Jan25	3106	36	12	$104	1.15%	
2	2025-01-03	FB_Jan25	3105	26	8	$102	0.84%	
3	2025-01-04	FB_Jan25	1107	27	9	$71	2.45%	
4	2025-01-05	FB_Jan25	1317	15	7	$78	1.10%	

Comparing campaigns performance

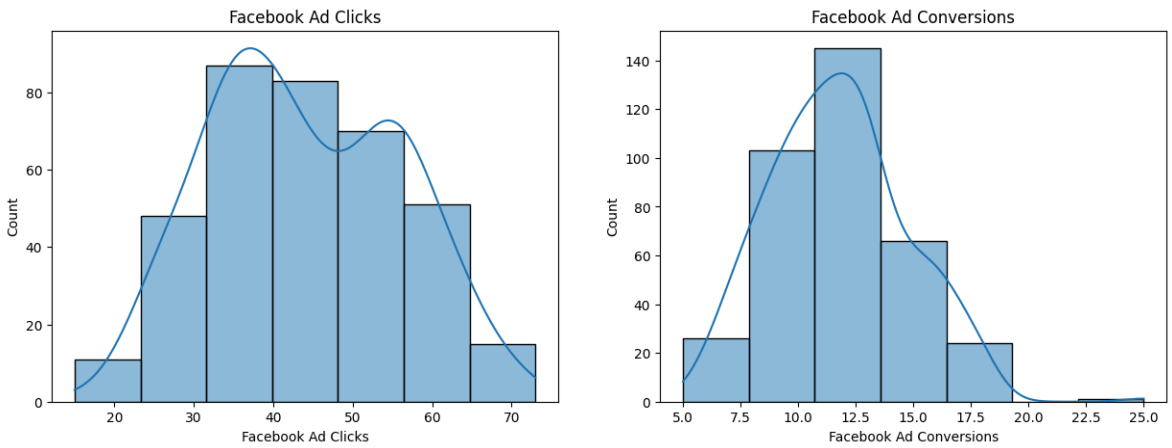
In [36]:

```
# distribution of clicks and conversions-
plt.figure(figsize=(15,5))

plt.subplot(1,2,1)
plt.title('Facebook Ad Clicks')
sns.histplot(df['Facebook Ad Clicks'],bins=7, edgecolor='k', kde=True)

plt.subplot(1,2,2)
plt.title('Facebook Ad Conversions')
sns.histplot(df['Facebook Ad Conversions'],bins=7,edgecolor='k', kde=True)

plt.show()
```



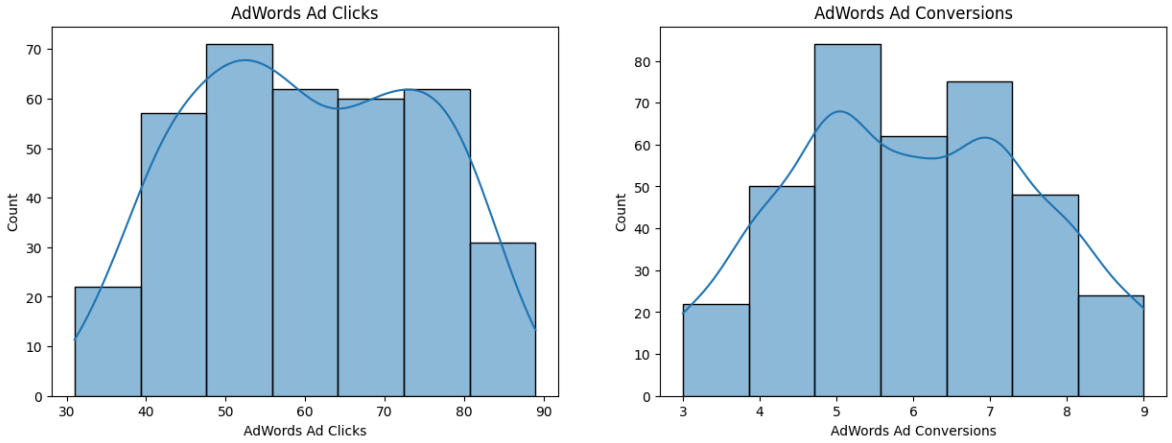
In [37]:

```
plt.figure(figsize=(15,5))

plt.subplot(1,2,1)
plt.title('AdWords Ad Clicks')
sns.histplot(df['AdWords Ad Clicks'],bins=7, edgecolor='k', kde=True)
```

```
plt.subplot(1,2,2)
plt.title('AdWords Ad Conversions')
sns.histplot(df['AdWords Ad Conversions'],bins=7,edgecolor='k', kde=True)

plt.show()
```



Frequently days with high and low numbers conversions:

In [53]:
```
# create function to calculate category for conversions -
def category_conversion(conversion_col):
    category=[]
    for conversion in df[conversion_col]:
        if conversion < 6:
            category.append('Less than 6')
        elif 6<= conversion < 12:
            category.append('6-11')
        elif 12<= conversion < 16:
            category.append('12-15')
        else:
            category.append('More than 16')
    return category

df['Facebook Conversion Category']= category_conversion('Facebook Ad Conversions')
df['AdWords Conversion Category']= category_conversion('AdWords Ad Conversions')
```

In [54]:
```
df['Facebook Conversion Category'].value_counts()
```

Out[54]:
```
Facebook Conversion Category
6-11           174
12-15          143
More than 16    47
Less than 6      1
Name: count, dtype: int64
```

In [55]:
```
facebook = pd.DataFrame(df['Facebook Conversion Category']).value_counts().reset_index().rer
```

In [56]:
```
facebook
```

Out[56]:

	Category	count
0	6-11	174
1	12-15	143
2	More than 16	47
3	Less than 6	1

In [57]:
```
adwords = pd.DataFrame(df['AdWords Conversion Category']).value_counts().reset_index().rena
```

In [58]:
```
adwords
```

Out[58]:

	Category	count
0	6-11	209
1	Less than 6	156

In [61]:
```
df.head()
```

Out[61]:

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Fa Con (Conv
0	2025-01-01	FB_Jan25	2116	18	8	$126	0.83%	
1	2025-01-02	FB_Jan25	3106	36	12	$104	1.15%	
2	2025-01-03	FB_Jan25	3105	26	8	$102	0.84%	
3	2025-01-04	FB_Jan25	1107	27	9	$71	2.45%	
4	2025-01-05	FB_Jan25	1317	15	7	$78	1.10%	

In [63]:
```
category_df = pd.merge(facebook,adwords, on = 'Category',how='outer').fillna(0)
```

In [64]:
```
category_df
```

Out[64]:

	Category	count_x	count_y
0	12-15	143	0.0
1	6-11	174	209.0
2	Less than 6	1	156.0
3	More than 16	47	0.0

In [65]:

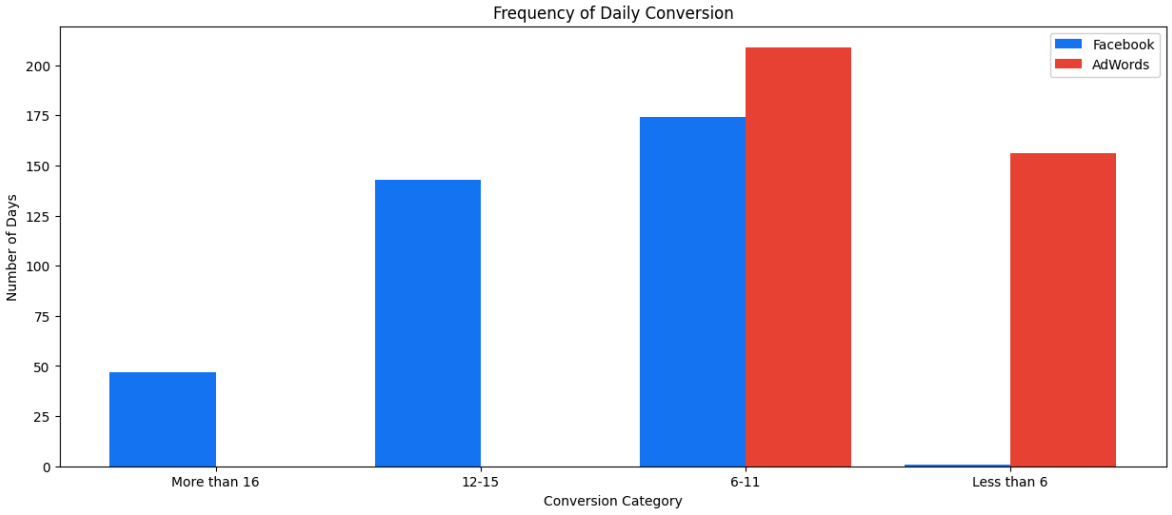
```
category_df= category_df.iloc[[3,0,1,2]]
category_df
```

Out[65]:

	Category	count_x	count_y
3	More than 16	47	0.0
0	12-15	143	0.0
1	6-11	174	209.0
2	Less than 6	1	156.0

In [71]:

```
X_axis = np.arange(len(category_df))
plt.figure(figsize=(15,6))
plt.bar(X_axis-0.2,category_df['count_x'],0.4,label = 'Facebook',linewidth=1,color='#1877F2')
plt.bar(X_axis+0.2,category_df['count_y'],0.4,label = 'AdWords',linewidth=1,color='#EA4335')
plt.xticks(X_axis,category_df['Category'])
plt.xlabel('Conversion Category')
plt.ylabel('Number of Days')
plt.title('Frequency of Daily Conversion')
plt.legend()
plt.show()
```

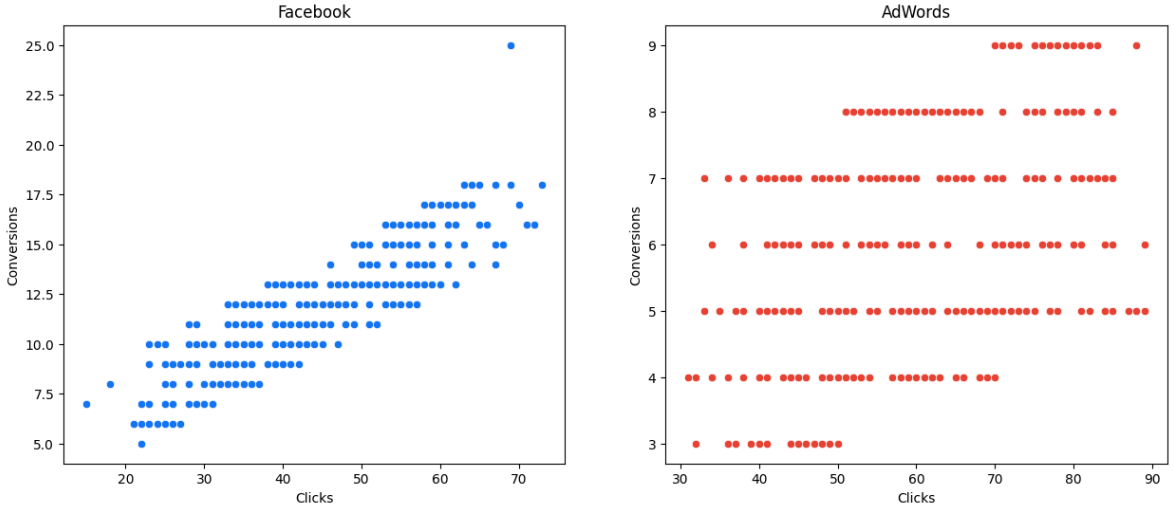


Do More clicks on the ad really lead to more sales-

In [72]:

```
plt.figure(figsize=(15,6))
plt.subplot(1,2,1)
plt.title('Facebook')
sns.scatterplot(x= df['Facebook Ad Clicks'],y=df['Facebook Ad Conversions'],color='#1877F2')
plt.xlabel('Clicks')
```

```
plt.ylabel('Conversions')
plt.subplot(1,2,2)
plt.title('AdWords')
sns.scatterplot(x= df['AdWords Ad Clicks'],y=df['AdWords Ad Conversions'],color='#EA4335')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.show()
```



In [73]:
```
facebook_corr = df[['Facebook Ad Conversions','Facebook Ad Clicks']].corr()
facebook_corr
```

Out[73]:

	Facebook Ad Conversions	Facebook Ad Clicks
Facebook Ad Conversions	1.000000	0.868808
Facebook Ad Clicks	0.868808	1.000000

In [74]:
```
adwords_corr = df[['AdWords Ad Conversions','AdWords Ad Clicks']].corr()
adwords_corr
```

Out[74]:

	AdWords Ad Conversions	AdWords Ad Clicks
AdWords Ad Conversions	1.000000	0.447993
AdWords Ad Clicks	0.447993	1.000000

Hypothesis Testing

Hypothesis: Facebook Ads greater number of conversion compared to AdWords Ads

Null Hypothesis: There is no differnce conversion between Facebook and AdWords

Facebook<=AdWords

Alternate Hypothesis: There is a differnce conversion between Facebook and AdWords

Facebook>AdWords

In [85]:
```
print('Mean Conversion \n')
print('Facebook:',round(df['Facebook Ad Conversions'].mean(),2))
print('AdWords:',round(df['AdWords Ad Conversions'].mean(),2))

t_stats, p_value = st.ttest_ind(a=df['Facebook Ad Conversions'],b=df['AdWords Ad Conversions']
print('TStatistic: ',t_stats,'\nP_value: ',p_value)

# significance value is 5%
if p_value <0.05:
   print('Reject Null Hypothesis')
else:
   print('Accept the Null Hypothesis')
```

```
Mean Conversion

Facebook: 11.76
AdWords: 5.98
TStatistic:  32.491588059013715
P_value:  3.018814514811081e-131
Reject Null Hypothesis
```

There is huge difference between Facebook Ads and AdWords Ads.

Facebook is higher than AdWords

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